The Spec Is the New Schematic — Speaker Notes (v3)

John Carter, N6YU — RSGB Convention 2026 · ~35 min talk + Q&A

1 The Spec Is the New Schematic

A spec does for an AI-built program what a schematic does for a radio

- A schematic lets someone somewhat skilled understand and recreate a project.
- A spec does the same for software: it guides a builder.
- A schematic is not a detailed project description with PCB layout, testing, alignment, and a BOM.
- Taking vibe coding to the next level.

2 Presentation Available on GitHub

Scan the QR code to follow along with the presentation, now or later.

- Viewing is optional: the QR code opens the presentation, so you can follow along or look at it later.
- The QR code opens the talk repo on GitHub; the slides PDF is linked at the top of the page.
- The talk repo, `github.com/jcarter-labs/rsgb-sdd-n6yu`, also has the Masterplan Generator. The link is repeated on the conclusion slide (17).

3 Software Is Replacing Solder

AI coding is like building an engineering prototype - not as good as off product line, but fully functional.

- Tools and test: the soldering iron and scope on the older side; modules and software on the newer side. Storage: the file cabinet becomes GitHub.
- For a sense of how big radio software has become, look at a modern logger: rig control, internet, display, database, callsign lookup, propagation, and more.
- There are ways to get around the drawbacks

4 Remote Operation in Mojave Desert

Like many of us, I am plagued with local noise, so I like to get away. Really far away.

- Shows a camp in the setup process, late in the day arriving, tent not up, sun going down. 17' vertical whip antenna.
- The map shows roughly where in the Mojave 350 mi from home, 120 mi from LA, close to nothing but the desert town "Boron".
- I have Starlink and solar out there, but no panadapter. Just an older but excellent rig, the KX3. I really like panadapters.

5 Why Did I Bother? A Poor Man's Panadapter

No existing app combined a band-scope, POTA spots, and spotter selection, so I built one.

- I could tap the I/Q outputs and build a 50 kHz scope with FFT and display software, but that is a lot of hardware. What if I just listen to local spotters? Isn't that like a panadapter?
- POTA.APP shows maps and spots for hunting, but its interface prioritizes the map, not a bandscope.
- N1MM on Windows shows a bandscope, but not POTA spots.

6 Possible SDD Projects for Radio

SDD fits almost any shack project, from platform ports to Arduino builds to station control.

- This is a laundry list. SDD suits nearly anything.
- The screenshot is a keyer built from K6GTE's, ported from Linux to Mac.

7 Vibe Coding, Can Be Amazing!

Vibe coding nails the first 20% but leaves guessed assumptions, brittle code, and a lost chat.

- POLL How many of you have vibe coded?
- POLL How many of you have used GitHub?

8 Key Terms

A small shared vocabulary is all you need to follow the rest of the talk.

- **Spec:** the AI community's name for a Masterplan, the whole document.
- **Masterplan (MP):** the 4-part Spec for your app: Ground Rules, Project Brief, Architecture, and Task Plan.
- **Masterplan Generator:** a document that guides you in writing a masterplan for your app. It makes complex software buildable without software-engineering skills.
- **SDD:** Spec-Driven Development, a process. The spec stays the source of truth as the code changes.

9 A Better Alternative: SDD

The model is better at extracting requirements from you than you are at defining them.

- Vibing wasn't working for organizations. SDD is not rigid, comes in many forms, and is still developing.
- The process uses AI to extract requirements: the model is better at extracting requirements from you than you are at defining them.
- Overall time is shorter: more time on the spec, less on coding and debugging. It is also self-documenting.

10 Preliminaries: Claude Code & GitHub Setup

Claude is vastly more powerful than chat.

- Claude Code is now available in Claude Desktop (little toggle in upper left)
- **GitHub is a Superpower.** Claude can help you figure it out.

11 SDD in a Nutshell: Masterplan Generator + Idea + Screenshot

This is the talk in one slide.

- The name is not special; call it anything. The contents are not unique either. The four sections come from best practices at many tech companies (Amazon, Anthropic, Microsoft).

12 Use Generator + Your Idea to Create a Masterplan

Ground Rules, Project Brief, Architecture, Task Plan: four steps from idea to a buildable plan.

- **Ground Rules (how we work): another Superpower.** Use the Generator's prompts as-is. They keep you out of vibing trouble.
- **Project Brief (what it does):** give Claude your app idea, and the Generator asks you questions to customize the Project Brief. This is where the unknown unknowns come out.
- **Architecture (built with):** If you don't know the answers, use Claude Desktop (or another Claude Code instance) to help fill it in.
- **Task Plan (in what order):** the Generator structures the project into about 5 chunks: Setup, Connections, Core Logic, Main Features, UI.

13 Example: Idea + Screenshot → Masterplan Generator

A short idea plus one screenshot is enough input; the generator interviews you for the rest.

14 The Masterplan Generator

The generator lives on GitHub as plain, readable text anyone can open and reuse.

- Each prompt carries a label: RULE (follow it during the build), DRAFT (write this part now and show me), USER INPUT (ask me first).

15 Claude Code with Claude Desktop as Helper (Screenshot)

Superpower: when Code's output is confusing, ask Desktop to explain and paste the answer back.

16 It Works! Multiplatform Screenshots

One spec produced the same working app on Mac, Windows, and Linux.

17 Conclusion

Vibe coding gets you started; the masterplan gets you finished, so start small with your own idea.

- Call back to the opening: like a schematic, the spec is what you keep and share.
- The QR code points to the GitHub repo for this talk, which has the Masterplan Generator and the presentation. Your first masterplan starts here.

18 Time for Your Road Trip! Your Idea (Questions?)

Close on the road-trip image and open the floor: Claude Code and GitHub are the transportation, your idea is the trip.

- Callback to slide 4: same desert trails. The idea is the destination, Claude Code and GitHub are the vehicle, and the spec is the plan you drive from.
- Invite questions. Leave this slide up through Q&A.
- Keep the slide numbers handy. Type a number to jump: **14** for the generator, **17** for the repo link, **19** for references.

APPENDIX

19 Appendix – References

> Where to go deeper on SDD and Claude Code.

20 Find out more… (closing)

> How to reach me.

- John Carter, N6YU, john@n6yu.com, `github.com/jcarter-labs`.